Chapter 3: A Story of Numbers (Rational Numbers)

Worksheet 3: Rational Numbers – Properties and Operations

Learning Objective: Students will understand rational numbers, their properties (closure, commutativity, associativity, distributivity); add, subtract, multiply, and divide rational numbers; represent rational numbers on a number line; and find rational numbers between two given rational numbers.

Worked Example

Add: $\dfrac{-3}{4} + \dfrac{5}{6}$
Solution: LCM(4, 6) = 12
$= \dfrac{-9}{12} + \dfrac{10}{12} = \dfrac{1}{12}$

Find a rational number between $\dfrac{1}{3}$ and $\dfrac{1}{2}$:
Mean $= \dfrac{1}{2}\left(\dfrac{1}{3} + \dfrac{1}{2}\right) = \dfrac{1}{2} \times \dfrac{5}{6} = \dfrac{5}{12}$

Questions 1–5: Easy / Recall

Q1. Write the additive inverse of: $\dfrac{-5}{7}$, $\dfrac{3}{8}$, 0.

Q2. Write the multiplicative inverse of: $\dfrac{-4}{9}$, 7, $\dfrac{-1}{5}$.

Q3. Is $\dfrac{-6}{-9}$ a rational number? Simplify it.

Q4. Arrange in ascending order: $\dfrac{-2}{3}$, $\dfrac{1}{4}$, $\dfrac{-1}{2}$, 0.

Q5. Find three rational numbers between $\frac{1}{5}$ and $\frac{2}{5}$.

Questions 6–10: Basic Application

Q6. $\frac{-5}{8} + \frac{3}{4} =$

Q7. $\frac{7}{12} - \frac{-3}{8} =$

Q8. $\frac{-4}{9} \times \frac{-3}{8} =$

Q9. $\frac{-7}{10} \div \frac{14}{5} =$

Q10. Verify: Is multiplication of rational numbers commutative? Use $\frac{2}{3}$ and $\frac{-5}{7}$.

Questions 11–15: Practice and Skill Building

Q11. Simplify: $\frac{-3}{4} \times \left(\frac{2}{5} + \frac{-7}{10}\right)$

Q12. The product of two rational numbers is $\frac{-9}{14}$. One is $\frac{3}{7}$. Find the other.

Q13. Represent $\frac{-5}{3}$ on a number line and find two rational numbers on either side of it. (Teacher Verification Required)

Q14. Verify the distributive property: $\frac{2}{5} \times \left(\frac{-3}{4} + \frac{1}{2}\right) = \frac{2}{5} \times \frac{-3}{4} + \frac{2}{5} \times \frac{1}{2}$

Q15. Using properties, simplify: $\frac{4}{7} \times \frac{-3}{8} + \frac{4}{7} \times \frac{-5}{8}$

Questions 16–18: Higher Order Thinking

Q16. Between $\frac{1}{7}$ and $\frac{1}{6}$, find 5 rational numbers using the formula $\frac{p}{q} = \frac{np}{nq}$.

Q17. The sum of two rational numbers is $\frac{-4}{5}$. One number is $\frac{7}{15}$. Find the other. Is it positive or negative?

Q18. Priya says: "The reciprocal of a negative rational number is always negative." Rahul says: "The sum of a rational number and its reciprocal is always greater than 2." Evaluate both statements with examples.

Questions 19–20: Real-Life Applications

Q19. A shopkeeper gains $\frac{1}{5}$ of the cost price on each item. If he sells a shirt for Rs. 960, find the cost price and the gain in rupees.

Q20. A tank is $\frac{3}{7}$ full of water. After using $\frac{1}{4}$ of the total tank capacity, $\frac{2}{3}$ of the remaining water is transferred to another tank. What fraction of the original full tank is now left in the first tank?

Reflection Question: Every integer is a rational number, but not every rational number is an integer. Explain with examples. Where do rational numbers fit among all types of numbers?